

26TH DIGITAL BLUE ECONOMY SUMMIT · EXPERIMENTAL PAPER

ParaSail

Predictive Advisory and Retrieval-Augmented System Advancing Informed Livelihoods

A geospatial decision support system for sustainable fishing, marine conservation and coastal livelihood resilience

Hard conservation constraints

Explainable by retrieval

Adaptable by LoRA fine-tuning

Grounded AI assistant (local)

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CONTENTS

What this deck covers

1

Problem, gap, objectives

The information gap is architectural, not algorithmic

2

Architecture and data

Six layers, open inputs, telemetry, resilient ingestion

3

The advisory engine

Dual-collection retrieval, scoring under hard constraints

4

Geospatial constraints

Ocean-masked zones and real protected-area boundaries

5

Grounded AI assistant

Local VLM, guardrails in code, honest availability

6

Validation

Six test families, gates G1-G6, end-to-end evidence

7

Deployment and impact

Hardware tiers, zero marginal cost, who benefits

8

Conclusion

The five summit questions in one pass

Tap any card to jump to that section; the CONTENTS link on each slide returns here.

The gap is architectural, not algorithmic

35.5%

of assessed stocks fished beyond sustainable levels

FAO SOFIA 2023

11 Mt

of plastic entering the ocean every year

UNEP 2023

~3×

projected growth of plastic flows by 2040

UNEP 2023

~492 M

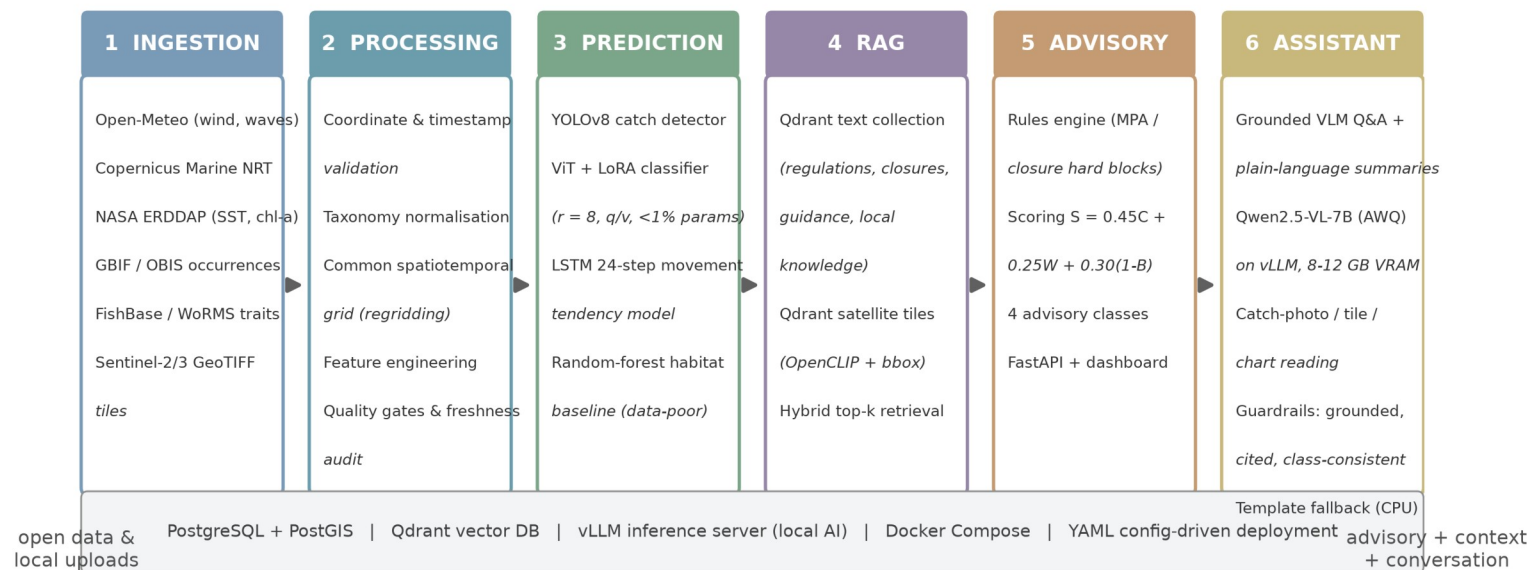
people supported by small-scale fisheries

FAO 2023

Tool class	Weather	Fish context	Legality	Explainability
Weather applications	✓	–	–	–
PFZ-style advisories	–	✓	–	–
Habitat / SDM studies	–	✓	–	–
Static regulations	–	–	✓	–
ParaSail (this work)	✓	✓	✓	✓

Objectives RO1-RO9: fuse live open data · model movement with habitat fallback · enforce MPAs and closures as hard constraints · attach regulations to every advisory · adapt locally via LoRA · expose API + dashboard · validate every layer · align with blue-economy principles · ground a conversational layer.

Six layers, one advisory



Every layer independently replaceable; region, species, closures and weights are YAML configuration, not code.

Station telemetry: 13 parameters in three groups (weather · sea · station health), fetched in independent groups so one unsupported variable degrades only its own group - and the same live set is what the assistant analyses.

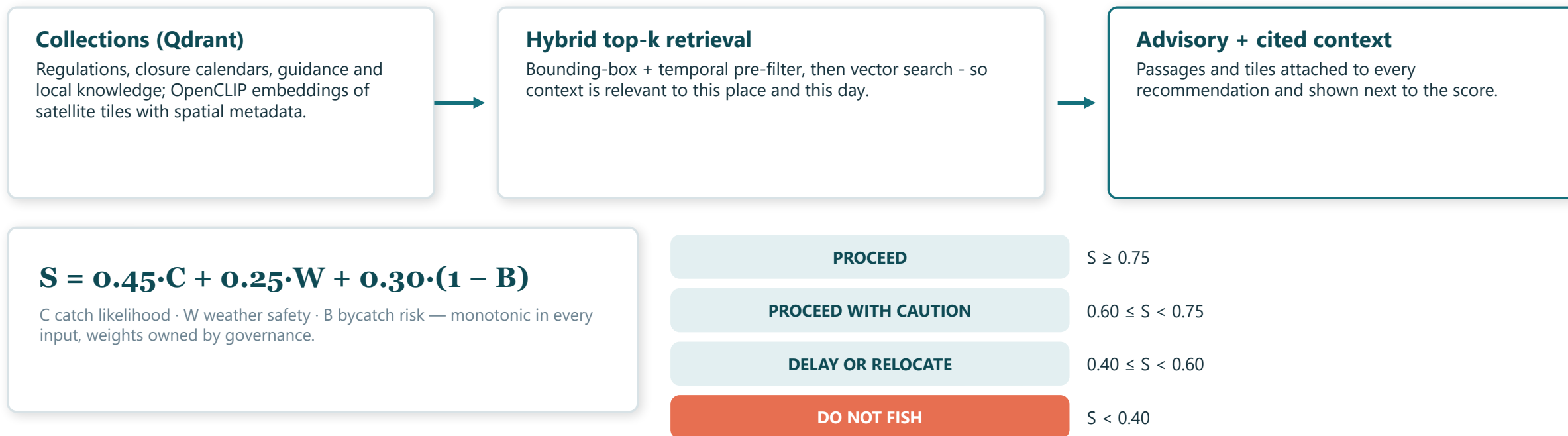
Open data in

- Open-Meteo — hourly met + marine forecasts
- Copernicus Marine — daily SST and currents
- NASA ERDDAP — SST, chlorophyll-a fields
- GBIF / OBIS — historical occurrences
- FishBase / WoRMS — taxonomy and traits
- Sentinel-2 / 3 — multispectral tiles

Resilience by design

- Freshness contracts per source
- Caching with explicit age flags
- Offline contingency — advisories continue from cached fields
- Quality gates, WoRMS normalisation

Every advisory carries its own justification

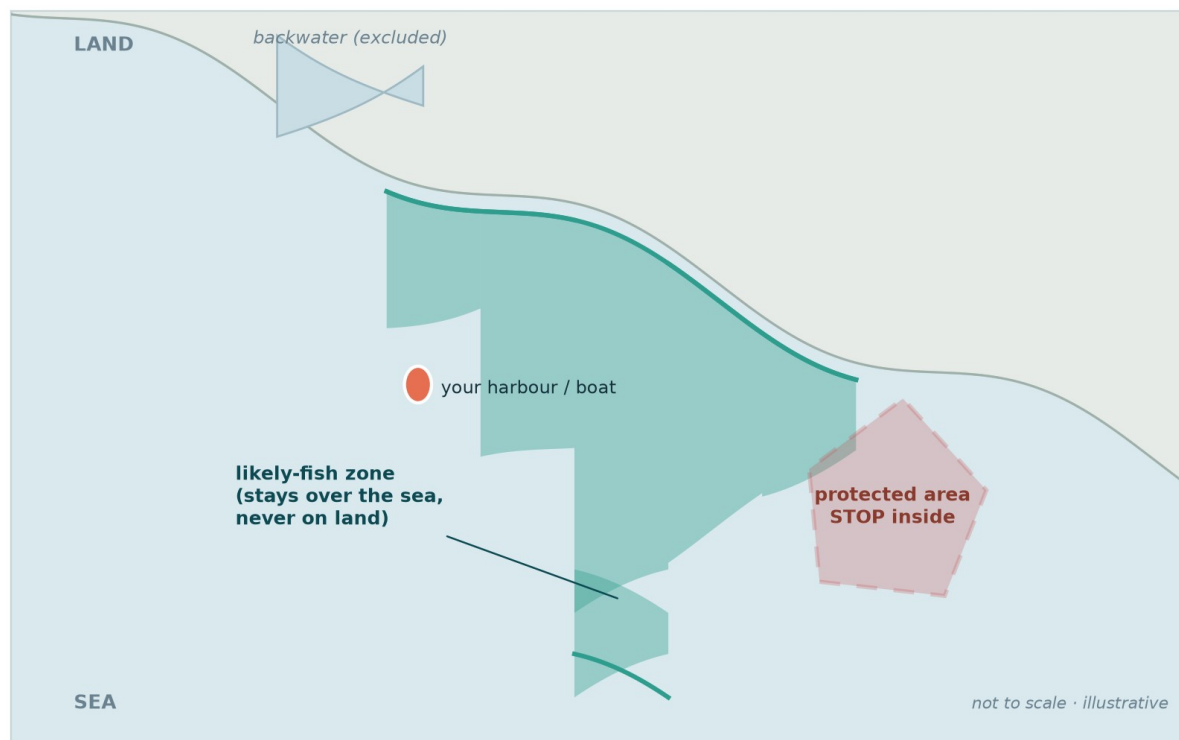


Hard rule: constraints run before scoring. Inside an active MPA or seasonal closure the advisory is DO NOT FISH regardless of score - no weight configuration can legalise an illegal trip.

Explainability is the retrieval of verifiable sources - not generated prose. New sources plug in as chunked documents; the corpus grows per region and no model is retrained.

Scalable by design: adding a regulation, a guidance note, a satellite pass or a whole regional corpus is an ingestion job - not a re-training project.

Constraints with real geometry



Illustrative schematic: blocky advisory cells clipped to the sea side, and a protected area that forces STOP.

1

Ocean mask (GSHHG)

Shoreline data unioned over the India box; a grid cell counts as sea only where the mask and the marine model agree. Lagoons and backwaters fall on the land side, so a zone can never sit over land.

2

Real protected boundaries

16 real Indian MPA polygons loaded into PostGIS. The guard is ST_Contains and fail-closed: an unreachable registry blocks the request instead of clearing it.

3

Fine grid, honest fallback

0.03° cells (about 3.3 km) traced into boundary polygons. The fallback chain is strictly sea-only, and if nothing qualifies there is no zone rather than a guess.

A grounded local assistant - it never decides

Inputs (grounding only)

Advisory JSON with live telemetry · top-k retrieved passages · optional image (catch photo, tile, chart) · question + language



Local VLM + registry

Qwen2.5-VL on vLLM (7B AWQ ≈ 6.9 GB, Apache 2.0) or Ollama (3B laptop tier). A runtime registry switches quality to hardware, 3B → 72B, probed for what is actually installed.



Guardrails (in code)

Citations must resolve to retrieved passages · the advisory class is authoritative · refusal over invention · backend named on every answer

Output: structured plain-language explanations (separate checkable points), cited answers, image readings - generated in English and pivoted through translation into ten coastal languages with vernacular fish names. If the GPU is down, a deterministic template assembles the same schema from the same data.

Structured output

Summaries return separate points, not a paragraph

Honest availability

Every registry model is probed; an uninstalled one is refused with the exact install command

Telemetry-fed

The live station set is part of the grounding payload

Translated, not guessed

One English generation, then a translation pivot - small models reason better in English

Validated end to end

T1 Live fetch



Live fetches reliable

Graceful degradation under outage, with explicit age flags

T2 Rule enforcement



Scoring monotonic

In every input; class thresholds land exactly as configured

T3 Scoring behaviour



Zones never over land

Ocean-masked zones stayed at sea at Kochi, Chennai and Mangaluru



Service hardened

Rate limits 429; 401/403 role gates; security headers on every response

T4 Retrieval relevance



Closure blocks exact

June sardine request returns DO NOT FISH with the reason



Retrieval relevant

Top-k context attached to every advisory on the dashboard

T5 Assistant behaviour



Assistant guarded

Citations resolve; verdicts hold; template fallback exact; availability honest



Pilot-ready

Installable PWA, offline shell, documented hardware tiers and cost model

T6 Security behaviour

Local AI on one GPU - no API bills

Tier	Hardware	Model / profile	Sustains
L · laptop	RTX 4050 Laptop, 6 GB	Qwen2.5-VL-3B, Q4 · 6 k ctx · 1-2 concurrent	Personal use, demos
1 · pilot	RTX 3060/4070, 12 GB	7B AWQ · 8 k ctx · 8 concurrent	Thousands of requests/day
2 · recommended	RTX 4090 / NVIDIA L4, 24 GB	7B AWQ · 16 k ctx · 32-64 concurrent	Tens of thousands/day
3 · regional	2-4× L4/A10G + LB, or A100 40 GB	7B AWQ replicas + semantic cache	Millions of requests/year

\$0 per question

Self-hosted inference: the marginal cost of an answer is electricity.

~\$16-55k/yr avoided

Per-token billing for 10k questions/day at commercial API rates.

~\$2k once + \$315/yr

The recommended 24 GB tier: one GPU plus power, owned outright.

Who benefits: fishers save fuel and avoid fines · the ocean gets its closures respected automatically · communities keep their knowledge in a lasting library · authorities get decisions anyone can audit.

Conservation, legality and safety are built into the objective function - ParaSail cannot recommend an illegal trip under any configuration.

Primary alignment: SDG 14 (Life Below Water) · supporting: SDG 2 · 8 · 9 · 13

The five mandatory summit questions

Q1 · Problem No real-time, integrated, explainable advisory system for sustainable fishing in data-poor regions.

Q2 · Importance Livelihoods, food security, marine conservation and safety meet at one information gap.

Q3 · Technology Geospatial AI (YOLOv8, ViT + LoRA, LSTM, random forest), dual-collection RAG, a self-hosted Qwen2.5-VL assistant on vLLM with a runtime model registry, open live APIs, FastAPI + PostGIS + Qdrant; hardened with rate limiting and RBAC, shipped as a PWA.

Q4 · Evidence Open environmental, biodiversity and satellite data; regulation datasets; a working system validated by six test families across six gates.

Q5 · Contribution A production-grade platform issuing sustainability-scored, explainable, conversational advisories with conservation enforced as a hard constraint - on local AI, at zero marginal cost.

Thank you · Anantha Krishnan AS · Naipunnya School of Management, Cherthala, India · 26th Digital Blue Economy Summit